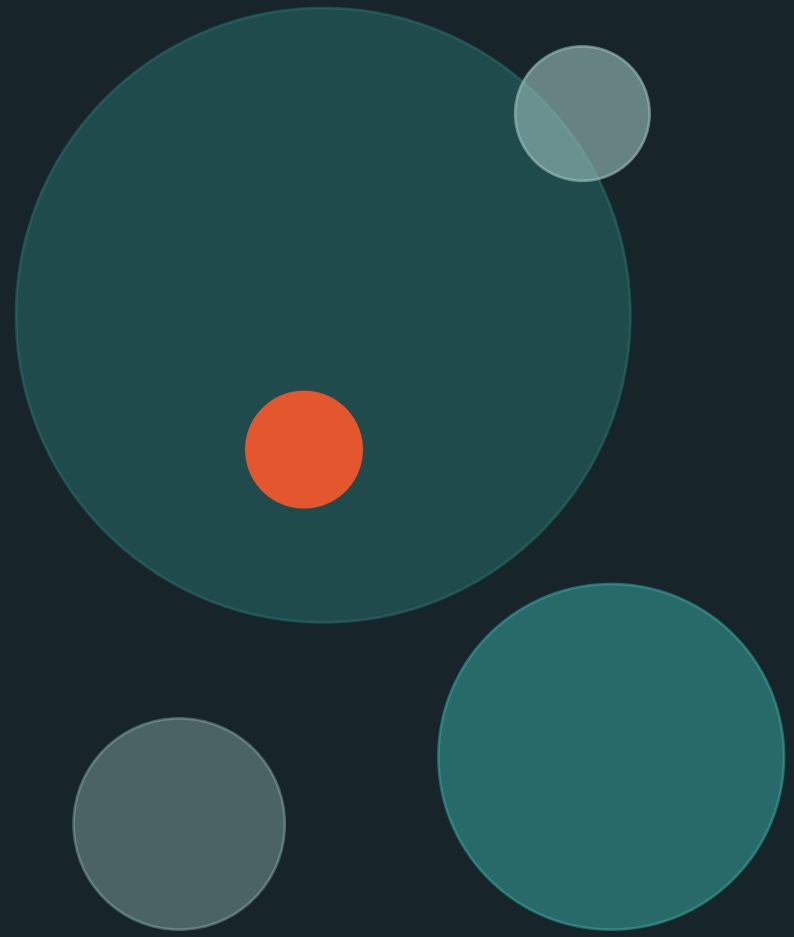
Fraud Detection: Intermediate Update

Machine Learning Project

Joshua Francis and Logan Digilormo



Where we left off



The problem

Flag fraudulent credit card transactions, which are a small share of all purchases.



The data

Simulated transactions from the public Sparkov generator on GitHub.



The plan

Four models: logistic regression, decision tree, random forest, gradient boosting.

Progress against our proposal

Generate public data	Done	822K transactions, 999 customers, 2025
Preprocess and engineer features	Done	31 inputs; history features use past data only
Time-based split	Done	Train Jan-Sep, validate Oct-Nov, test Dec
Train four models	Done	Tuned on validation, tested once
Metrics: precision, recall, F1	Done	Plus PR-AUC and confusion matrices
Demo, ablation polish, report	Next	First ablation run complete

Data: generated and validated

822,484

transactions

999

customers

1.2%

of transactions are fraud

98.8%

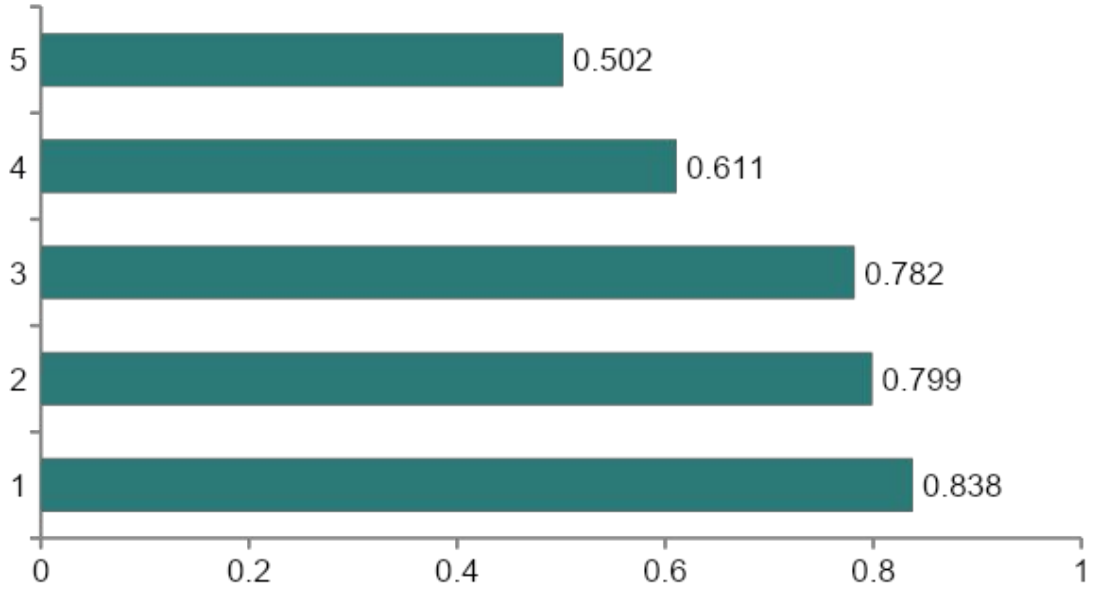
accuracy of always guessing "not fraud". It catches zero fraud.



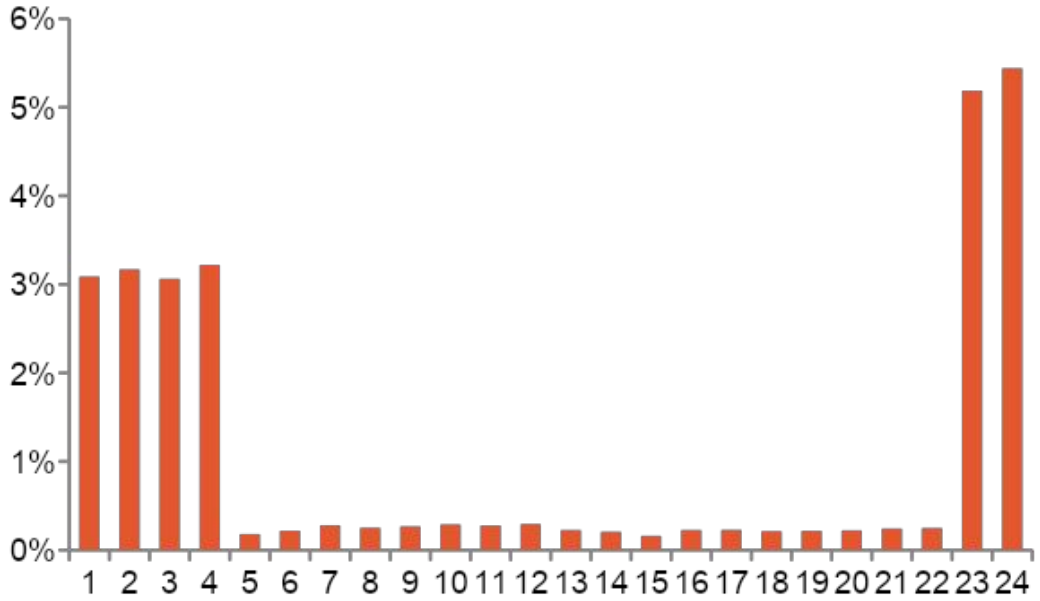
No private data: everything is simulated, and the generator is public and reproducible.

What the data shows

How well one feature separates fraud (1.0 is perfect, 0.5 is a coin flip)



Fraud rate by hour of day



Distance has no signal here: a surprise we report as a finding, not hide.

Features built so far



Base and category

Amount, age, city size, and 14 category columns.



Time

Hour, weekday, weekend flag, and night flag.



Distance

Haversine miles from home to the merchant.



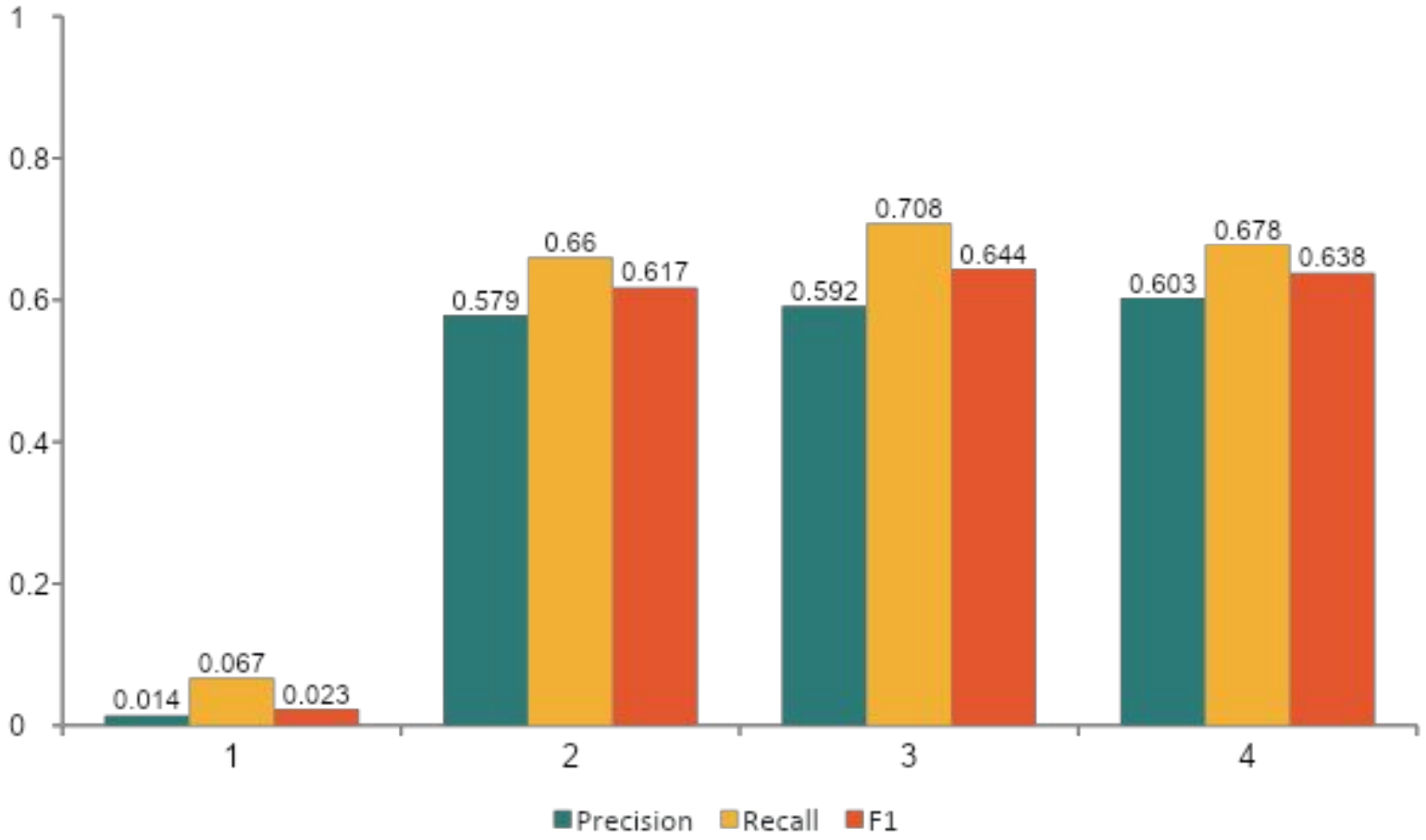
History

9 features: gaps, speed, amount vs average, counts.



No future leakage: history features only use a customer's earlier transactions.

Early model results



0.73

PR-AUC for the random forest, our best model so far. Logistic regression scores 0.004.

Limitations

Simulated data

Results may not transfer to real-world spending.

No distance signal

Score 0.50; adding it did not help in our first ablation.

Fraud rate drifts

1.3% in training versus 0.5% in the test month.

Light tuning so far

Small grids; gradient boosting stopped early at 34 trees.

Scope and topic: any changes

Topic and title

Unchanged: credit card fraud detection.

Changes since proposal

Four models instead of two, a time-based split, Sparkov data from GitHub.

Shift in emphasis

Amount, time, and customer history lead; distance is reported honestly.

Next steps and who does what



Josh

Models: Logistic regression and Decision tree

Also: Feature pipeline and the ablation study



Logan

Models: Random forest and Gradient boosting

Also: Deeper tuning and threshold choice



Together: scoring demo, final deck (Nov 22), report (Dec 6), and the GitHub repo.